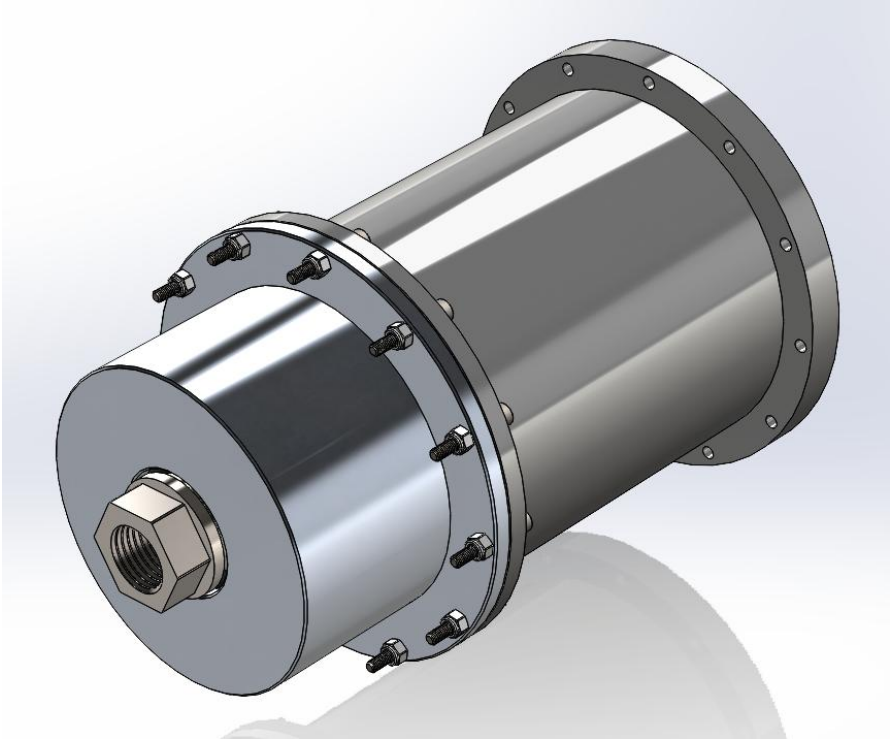
Engineering Design of Actuated Sensor Alignment Systems for Subsea Applications



Alistair Murison

MEng Mechanical Engineering Final Year Project Presentation

What is the Problem?

- Accurate flow data required for tidal site selection and annual energy estimations
- Velocity measurement devices are Acoustic Doppler Current Profilers (ADCP)
- ADCPs sit in a passively aligned gimbal frame
- Strong tidal forces knock the apparatus around
- Data quality is affected as a result
- ADCP compass requires calibration
- No commercial actuators suitable



Gravity mooring during deployment


Existing passively aligning gimbal ring system

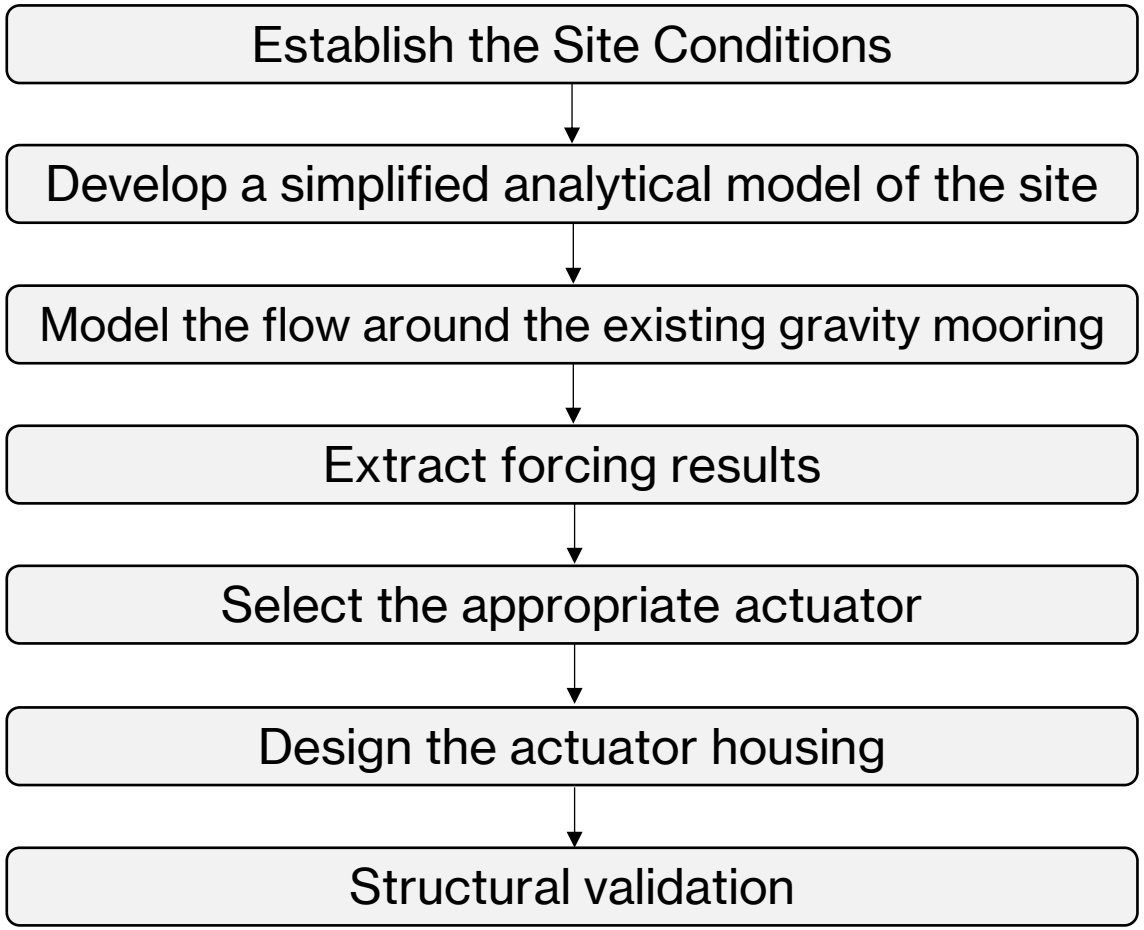
Design Brief

- Low-cost but high-performance actuator
- Ability to sustain long deployment
- 100 m depth rating
- Ability to hold position unpowered
- Fit into existing gimbal ring concept

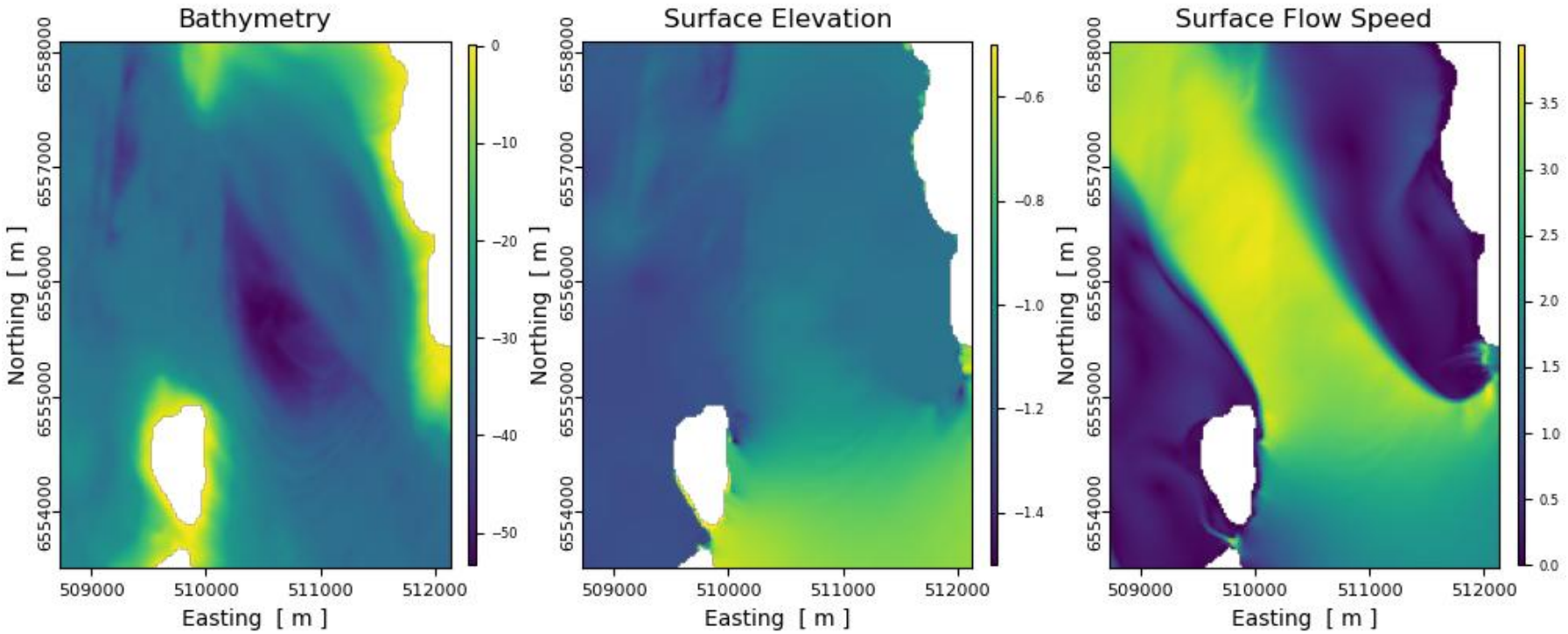


Sidus SS350 Heavy Duty Subsea Rotator [2]

The Plan to Meet the Brief



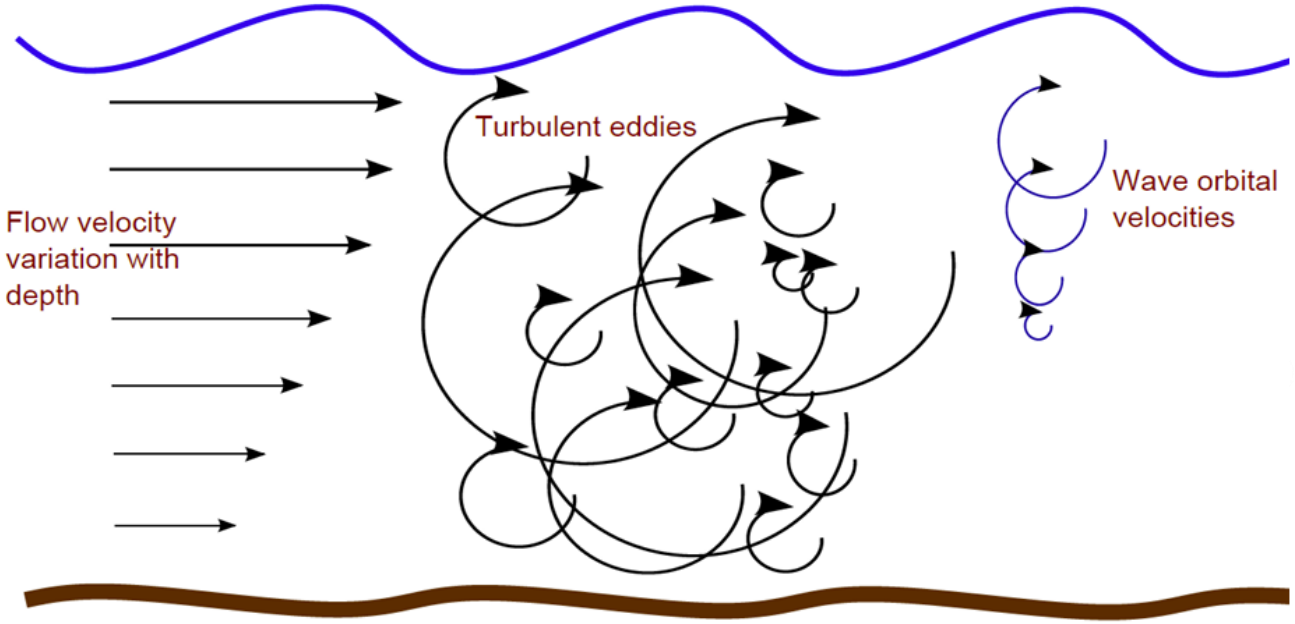
The Site – Fall of Warness, Orkney



Physical characteristics of the site [4]

- Depths of up to 50 metres
- Flow speeds up to 3.9 metres per second during spring tides
- Wave heights up to 3.9 metres in extreme conditions
- Turbulence Intensity 9 ± 1.9%

Analytical Flow Model



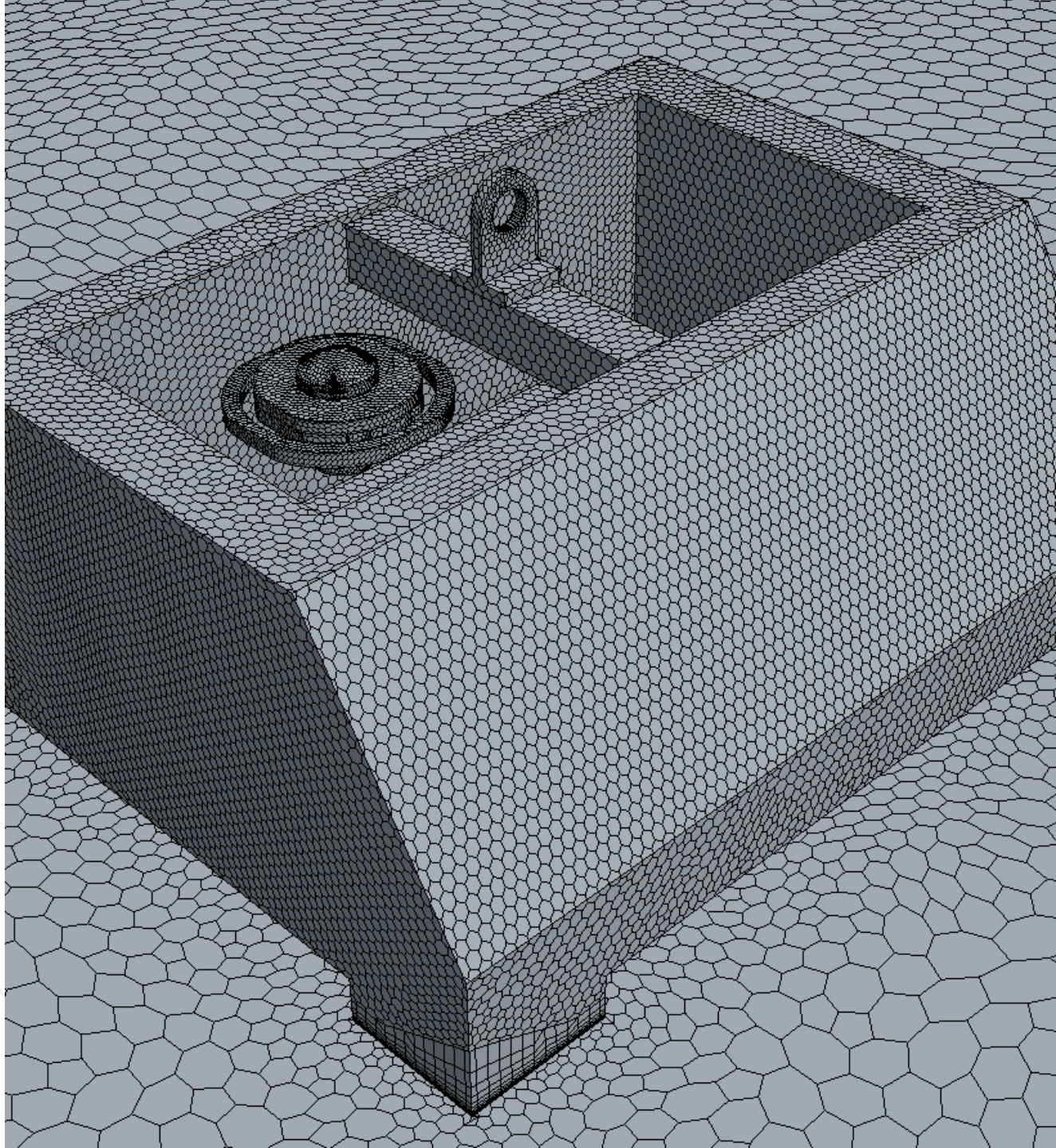
The unsteady characteristics of turbulent flow [5]

- Contributors to flow:
  - Current
  - Wave orbital velocities
  - Turbulent eddies

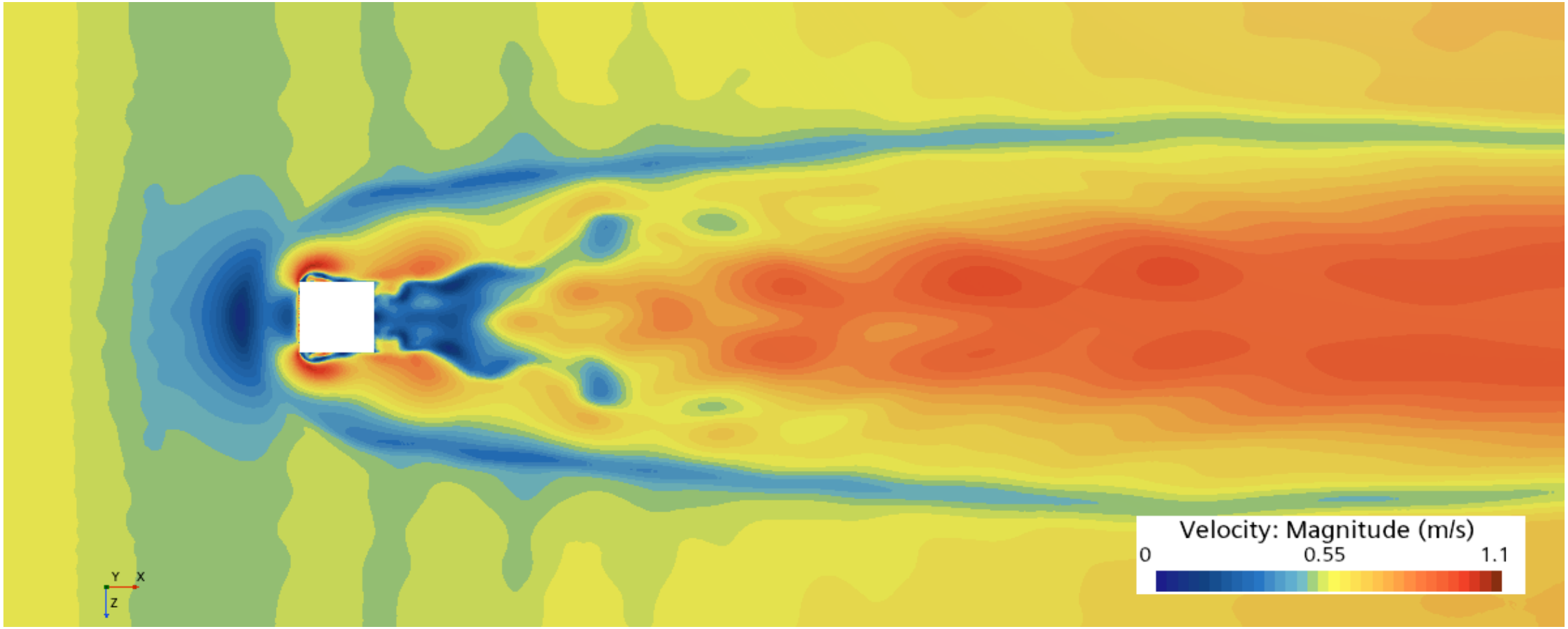
- Current modelled using von Karman (log-law) profile
- Waves modelled using Stokes 2nd order wave theory
- Model used as an inlet condition for flow modelling

Flow Modelling

- STAR-CCM+ Computational Fluid Dynamics (CFD) used
- Existing frame, gimbal set, and ADCP used
- Monitor plots used to measure forces on gimbals and payload over wave periods
- Expected bluff-body flow externally
- Expected recirculation within the frame

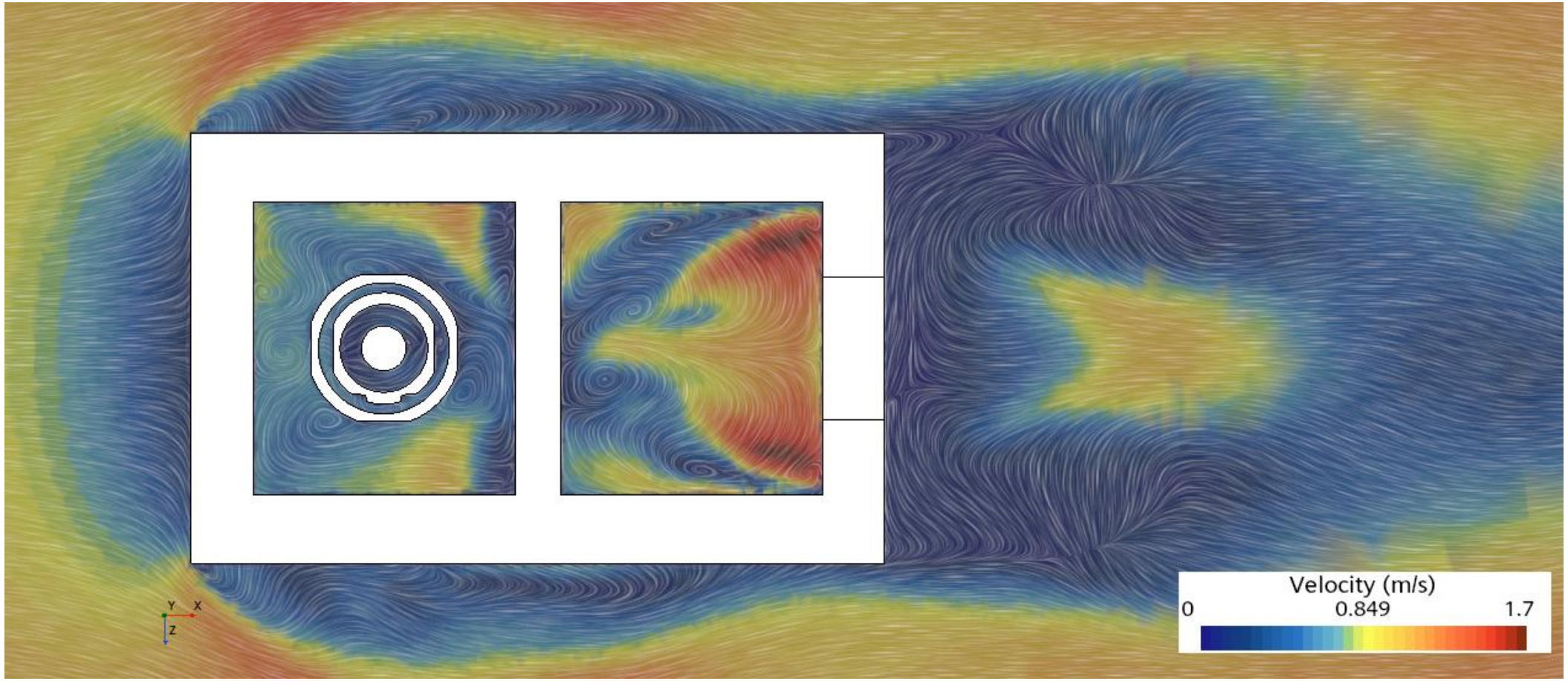


Flow Modelling



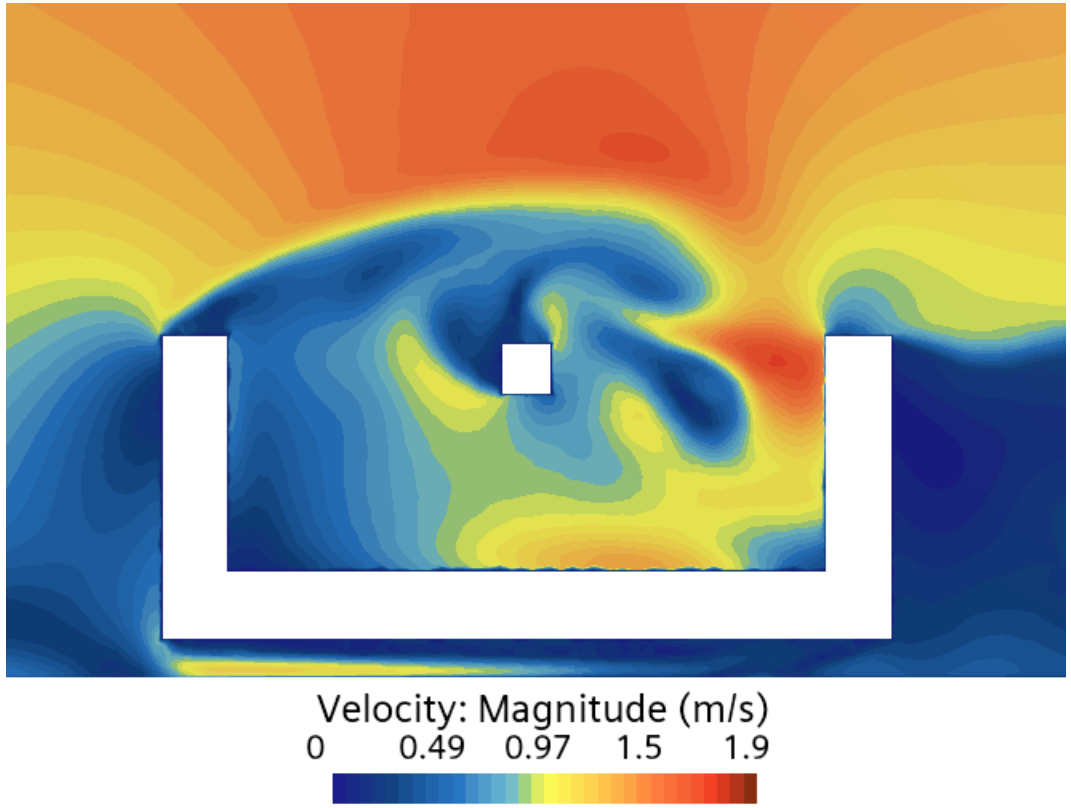
Horizontal velocity magnitude scalar plot – height 0.1 metres from seabed

Flow Modelling

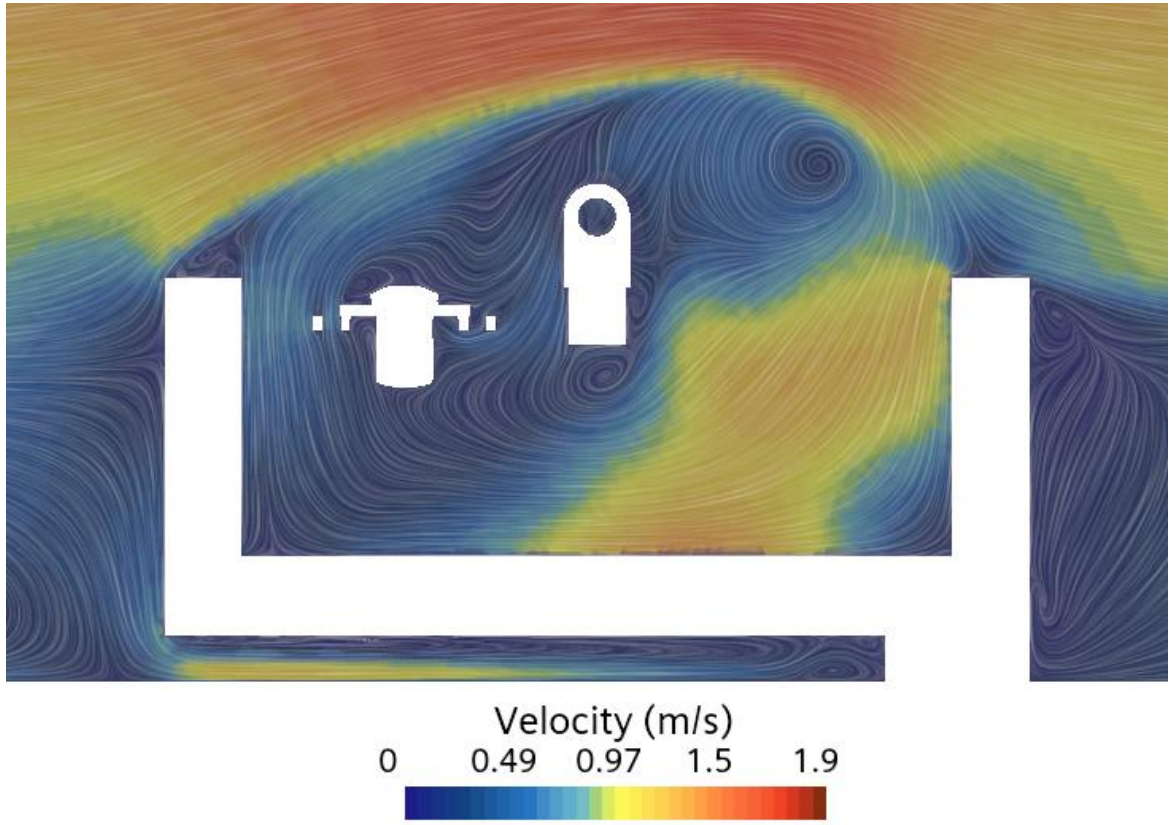


Horizontal velocity magnitude vector plot – height 0.75 metres from the seabed

Flow Modelling

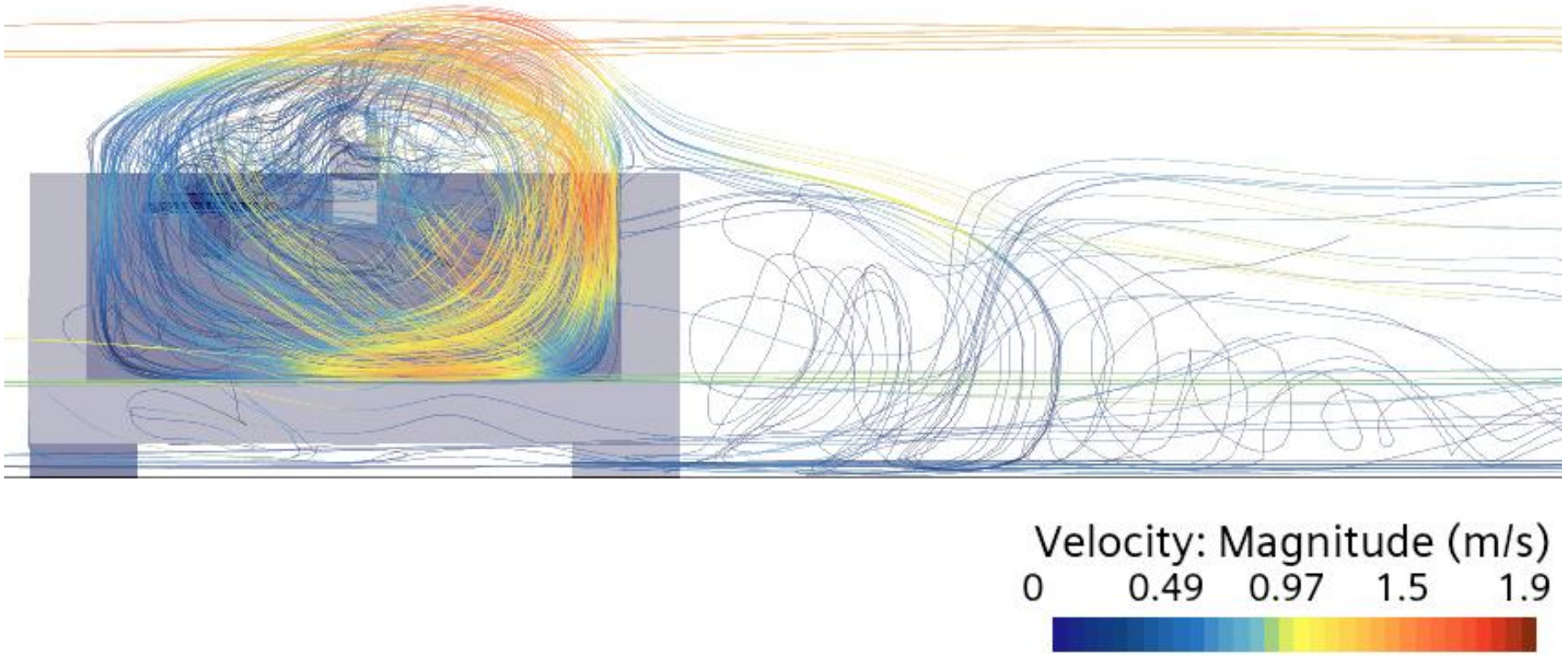


Vertical velocity magnitude scalar plot – 0.28 metres from centreline



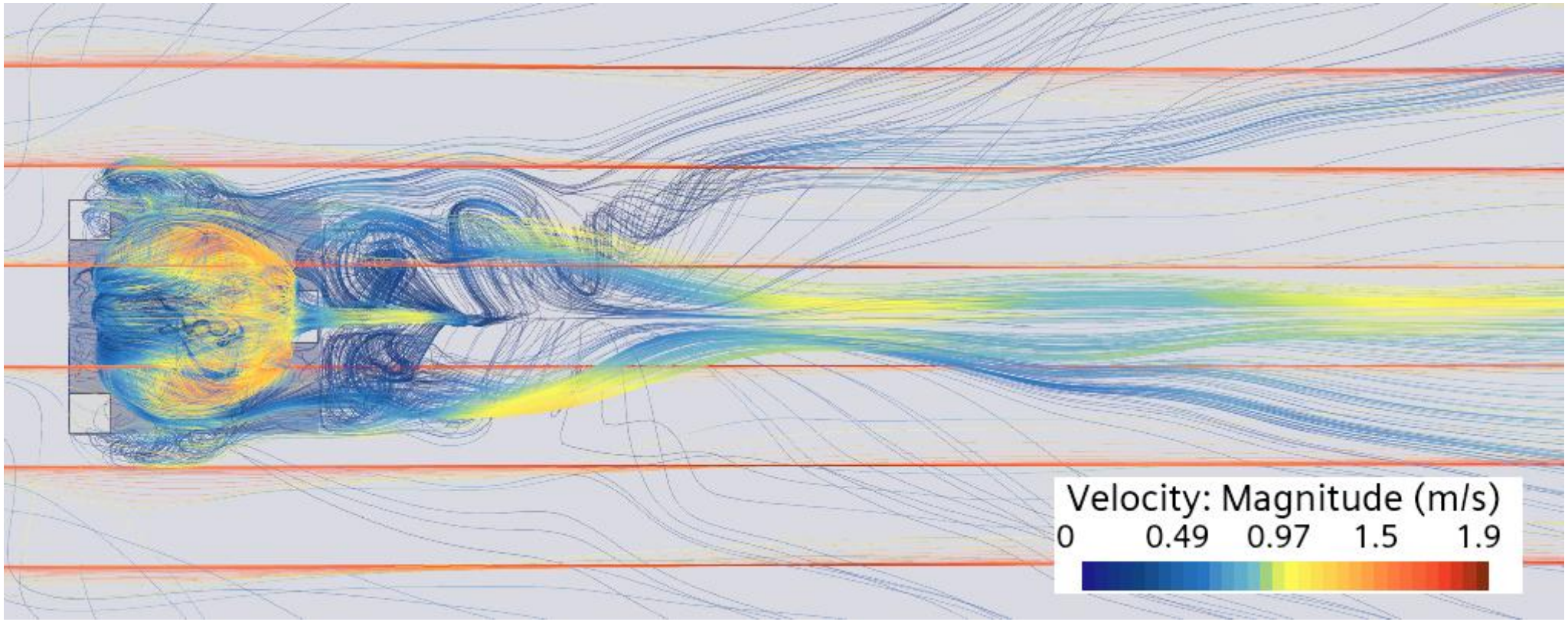
Vertical velocity magnitude vector plot – along centreline

Flow Modelling



Streamline velocity magnitude scalar plot – shows the path of individual water particles

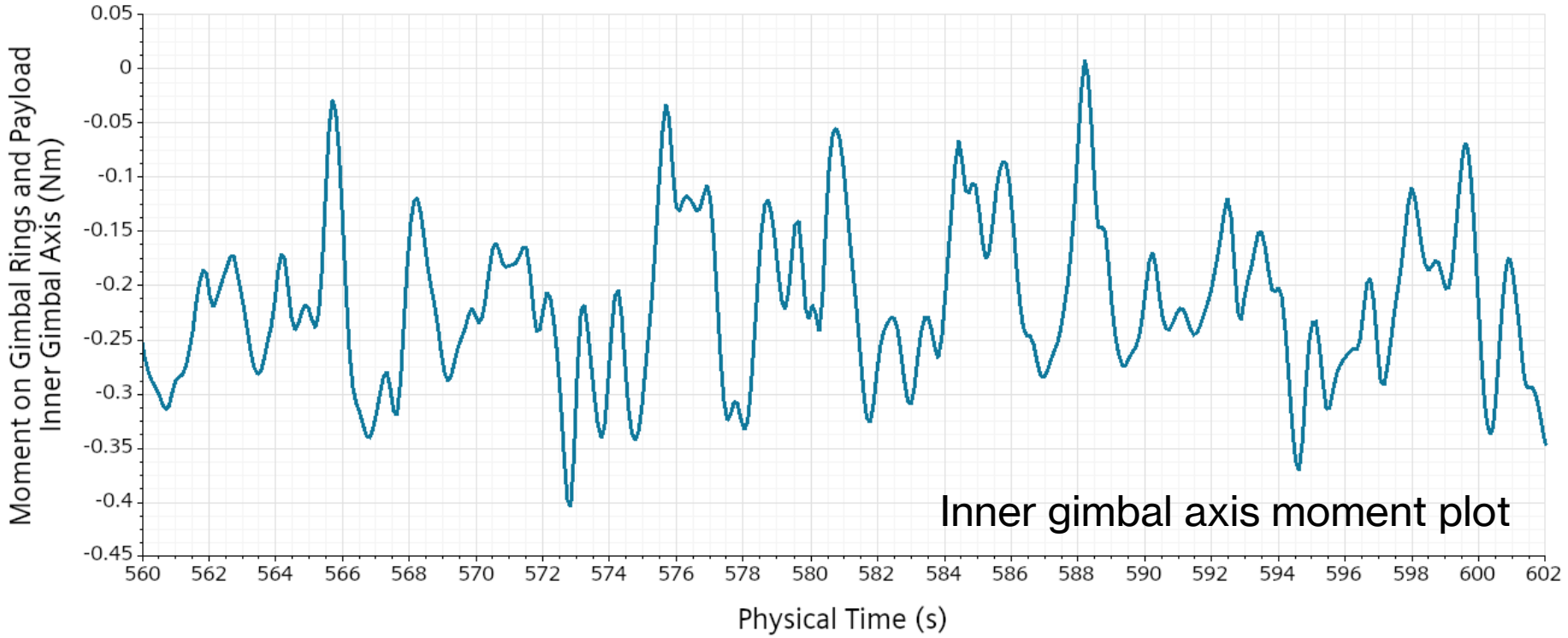
Flow Modelling



Streamline velocity magnitude scalar plot – shows the path of individual water particles

Key Results

- Maximum moment force on outer gimbal axis – **1.1851 Nm**
- Maximum moment force on inner gimbal axis – 0.4048 Nm



Inner gimbal axis moment plot

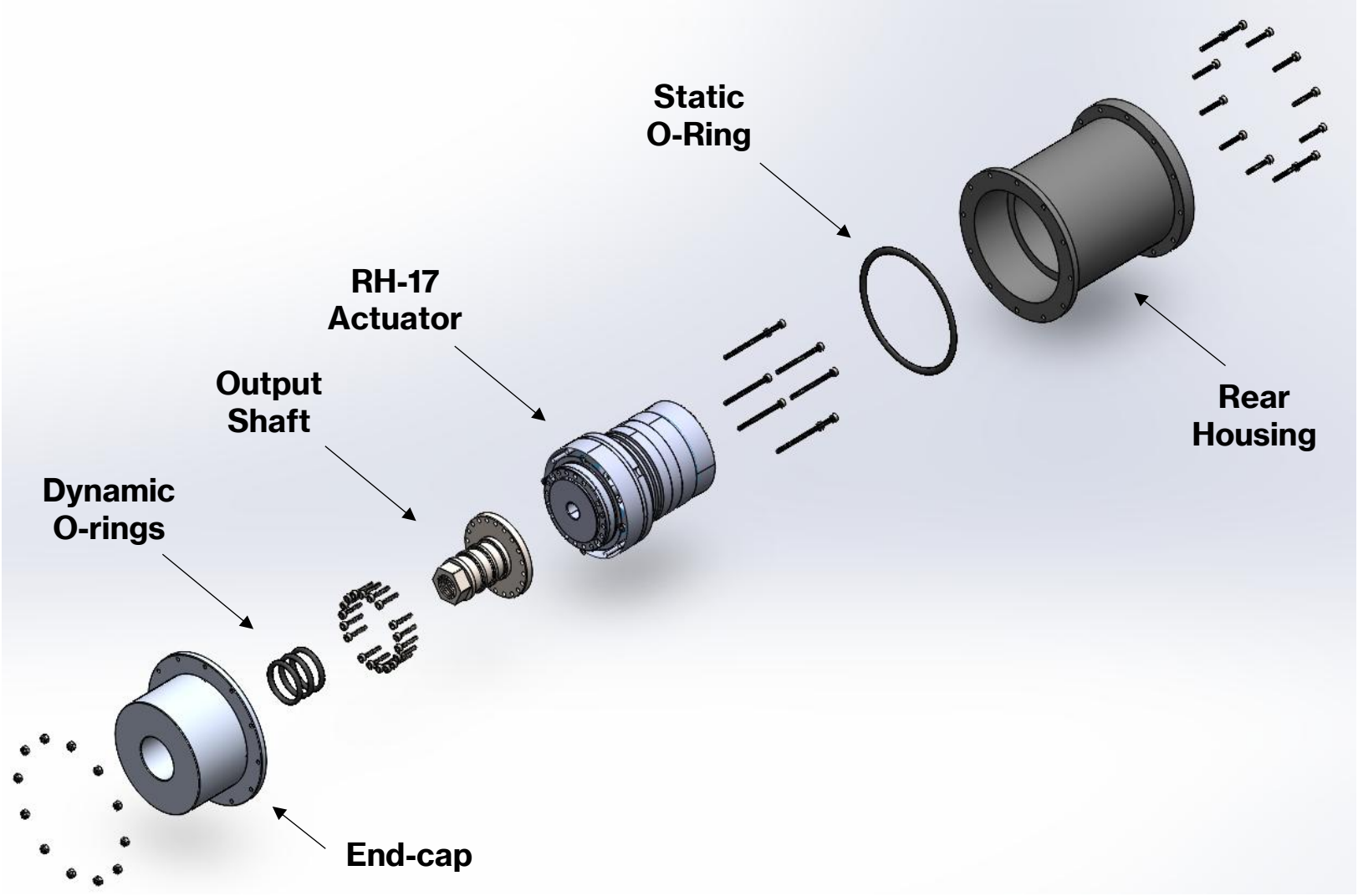
Chosen Actuator

- Chosen Actuator – MyActuator RH-17 Harmonic Series with Brake
- Required holding force – 1.1851 Nm
- Brake holding torque – 50 Nm
- Design load safety factor - 42

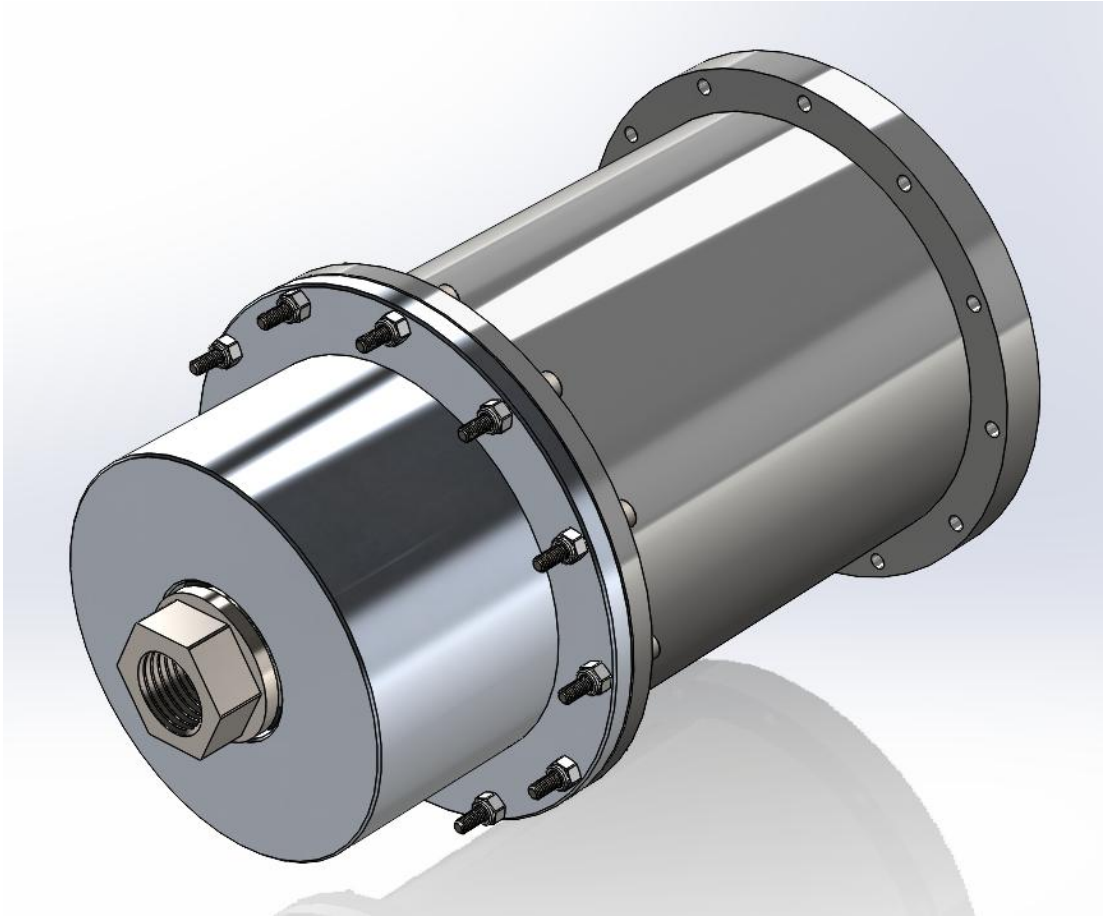


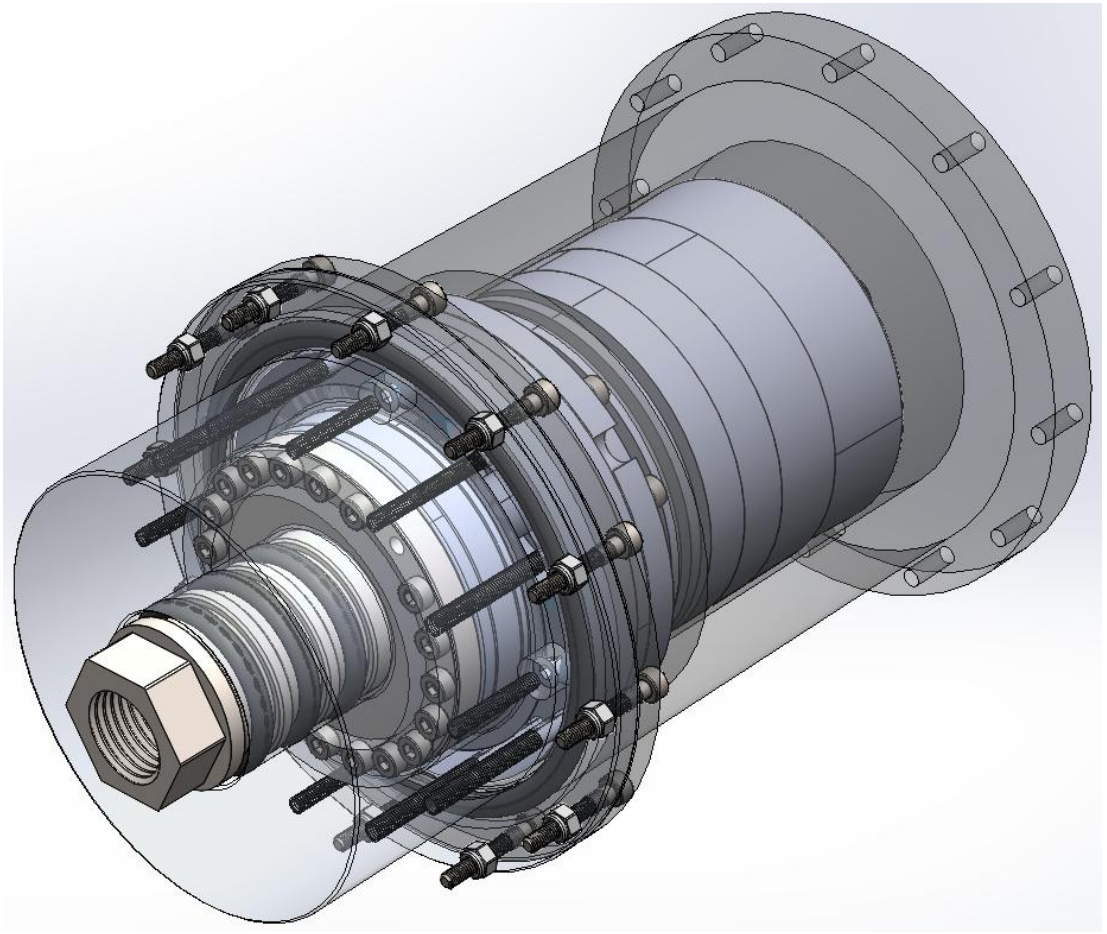
RH-17 actuator [6]

Design of Actuator Housing



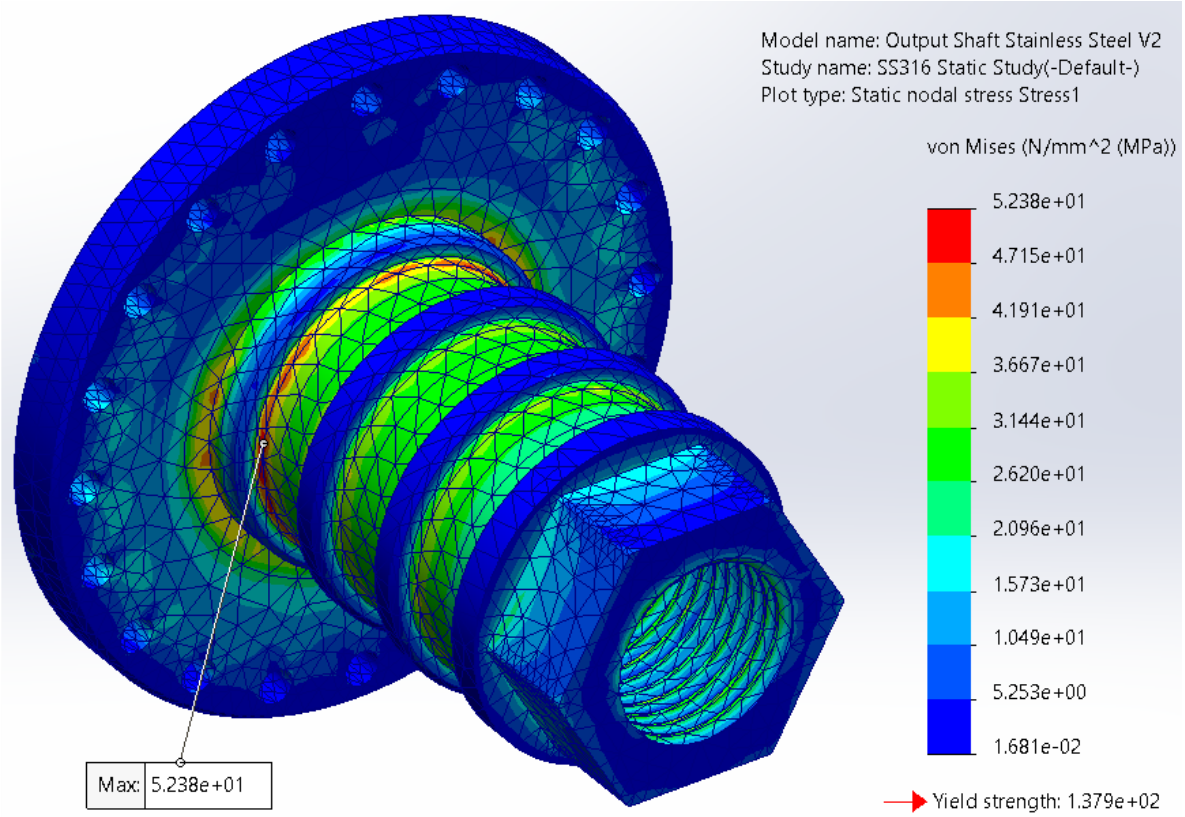
Design of Actuator Housing





Structural Analysis

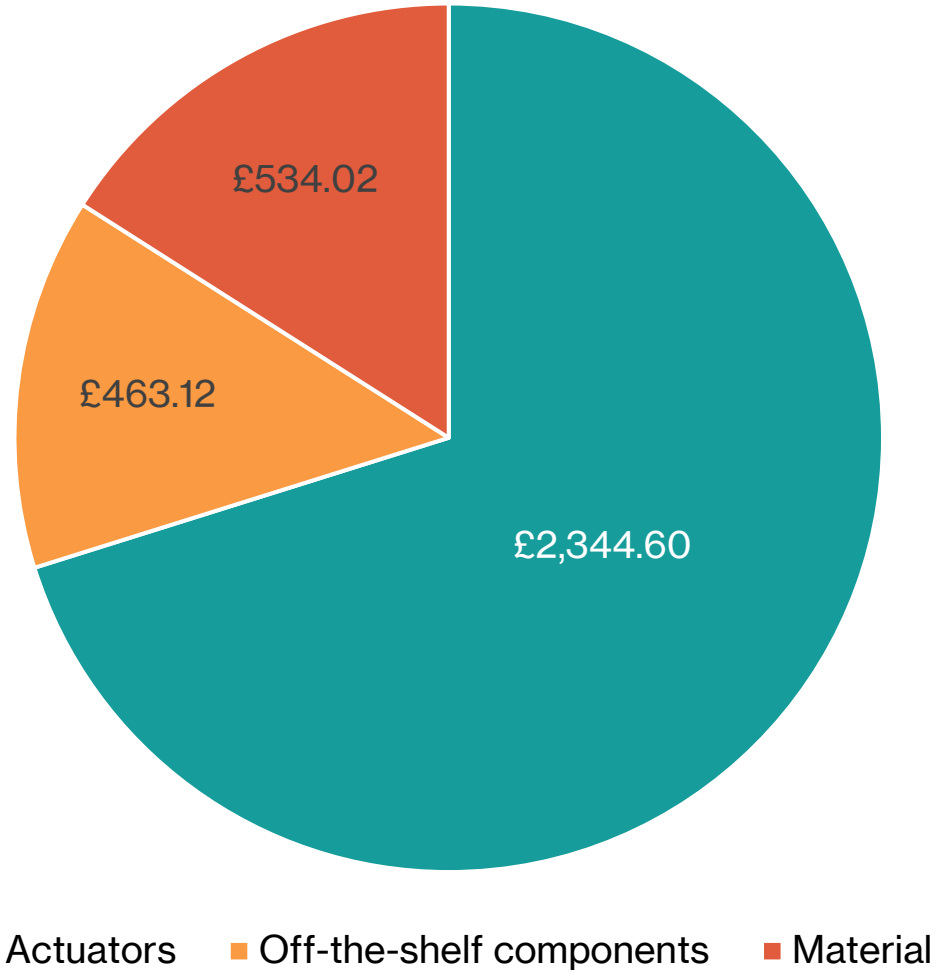
- Structural analysis performed using Finite Element Analysis
- Conservative loadings applied:
  - Pressure – 11.07 bar
  - Torque – 54 Nm
  - Remote Force – 77 N
- Components rated above Safety Factor of 2:
  - Output shaft – 2.63
  - End-cap – 5.9
  - Rear Housing – 2.6



Costs

- Total cost for pair: £3,341.75
  - Actuators: 70%
  - Off-the-shelf components: 14%
  - Material: 16%

- Cost saving of 63.4% (£5778.26) compared to commercial actuator pair
- Manufacturing cost not included



Key Outcomes

- Basic analytical model created for the Fall of Warness, Orkney
- CFD model created for future projects
- Suitable actuator chosen
- Housing prototype ready for manufacture and testing

Next Steps

- Further CFD work to reduce uncertainty in results
  - Reverse flow
  - Inclined seabed
  - Upright gimbal rings
- Manufacture and testing
  - Design of gimbal rings
  - Testing of sealing techniques
  - Operational performance
  - Control systems
  - Design iteration and frame optimisation

Q&A

Refences

[1] B. Sellar, Personal Communication, October 2025

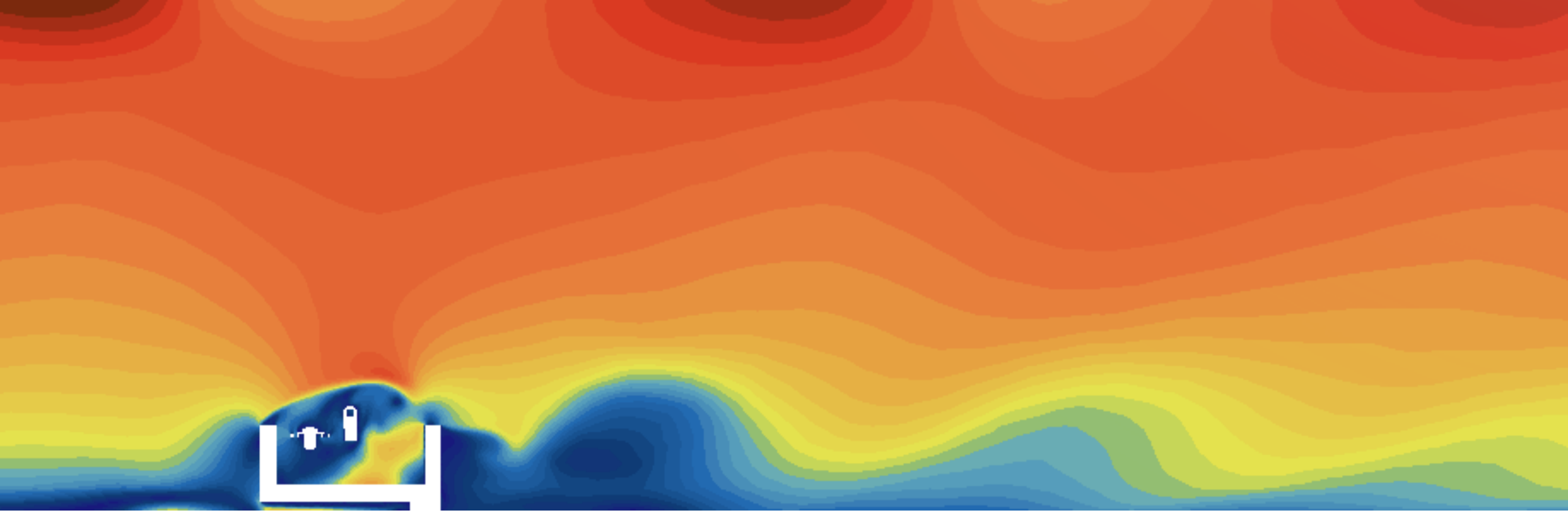
[2] SS350 Heavy Duty Subsea Rotator | Sidus Solutions. Accessed: 17th Mar. 2026. [Online]. Available: https://www.sidus-solutions.com/product/ss350-heavy-duty-rotator/.

[3] EMEC- Fall of Warness Wildlife Observation Programme | marine.gov.scot, 2015. Accessed: 17th Mar. 2026. [Online]. Available: https://marine.gov.scot/?q=information/emec-f all-warness-wildlife-observation-programme.

[4] A.Naberezhnykh, D. Ingram and I. Ashton, 'Wavelet applications for turbulence characterisation of real tidal flows measured with an ADCP,' Ocean Engineering, vol. 270, p. 113616, Feb. 2023, ISSN: 0029-8018. DOI: 10.1016/j.oceaneng.2022.113616. Accessed: 17th Mar. 2026. [Online].

[5] B. Sellar, C. Old, P. McCallum, N. Zigrat, M. Jordain De Thieulloy and M. Dorward, 'D2.2 Next Generation Flow Measurements and Flow Classification,' Tech. Rep., 2021. Accessed: 2nd Nov. 2025. [Online]. Available: https://tidalenergydata.org/media/documents /RLT-WP2-2-PDL-000-04.pdf.

[6] RH-17-D Details, en. Accessed: 23rd Mar. 2026. [Online]. Available: https://www.myactua tor.com/rh-17details.



Appendix

Flow Conditions and Analytical Equations

$$\mathbf{u}(z,t) = \underbrace{\mathbf{U}_c(z)}_{\text{current flow}} + \underbrace{\mathbf{u}_w(z)}_{\text{waves}} + \underbrace{\mathbf{u}'(z,t)}_{\text{turbulence}}$$

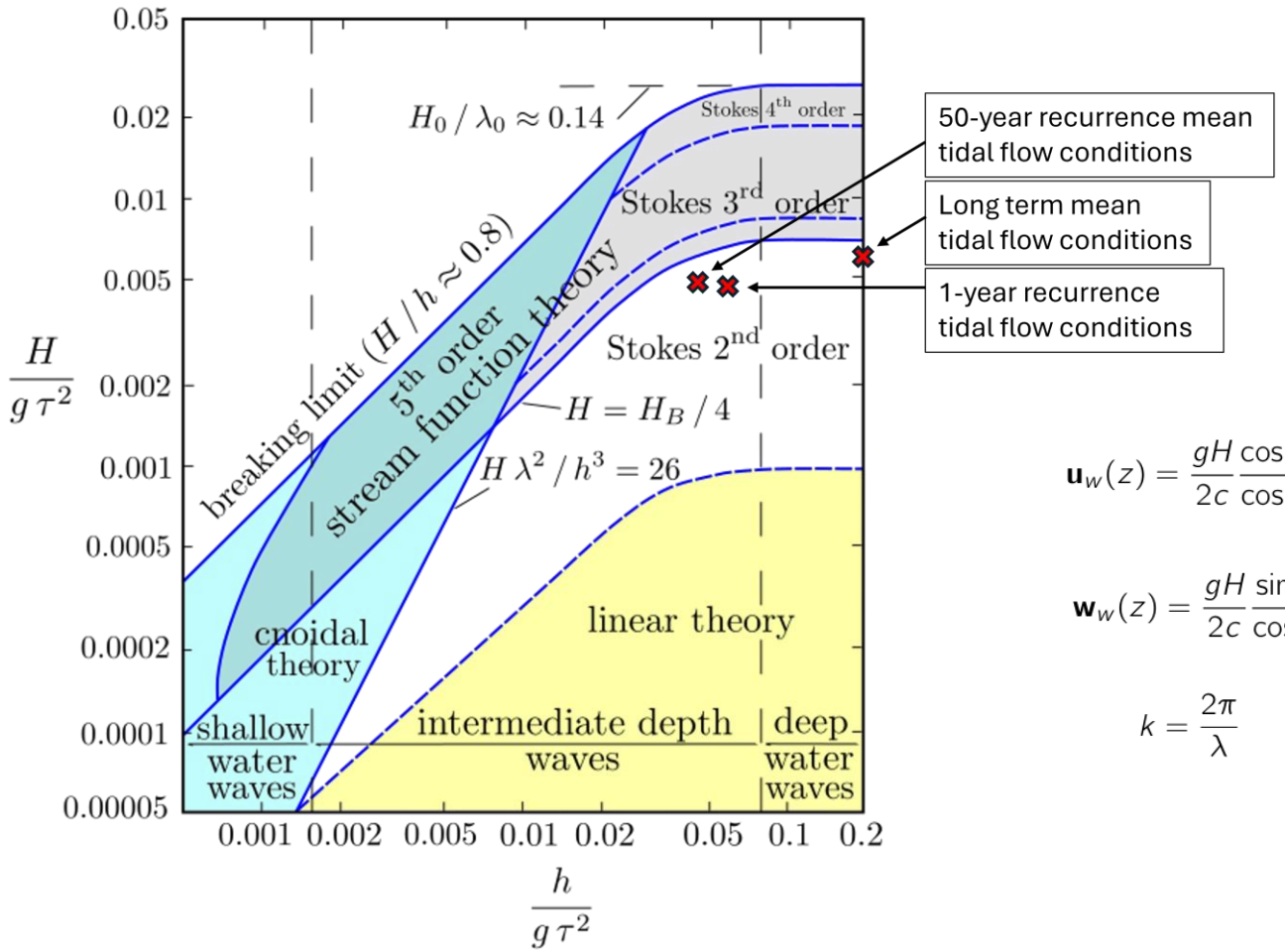


Table 3.1: Flow conditions values at FoW

	Long term mean	1-year recurrence	50-year recurrence
Current velocity [m/s]	2.420	3.911	4.212
Significant wave height [m]	1.11	3.9	5.2
Spectral Period [s]	4.2	9.1	10.4

$$\frac{\mathbf{U}_c(z)}{\mathbf{U}_c(z_1)} = \frac{\frac{u^*}{\kappa}\ln\left(\frac{z}{z_0}\right)}{\frac{u^*}{\kappa}\ln\left(\frac{z_1}{z_0}\right)} = \frac{\ln\left(\frac{z}{z_0}\right)}{\ln\left(\frac{z_1}{z_0}\right)} \quad \Rightarrow \quad \mathbf{U}_c(z) = \frac{\mathbf{U}_c(z_1)}{\ln\left(\frac{z_1}{z_0}\right)}\ln\left(\frac{z}{z_0}\right)$$

$$\Rightarrow \quad \mathbf{U}_c(z) = 0.459\ \ln\left(\frac{z}{0.01}\right)$$

$$\mathbf{u}_w(z) = \frac{gH}{2c}\frac{\cosh(kz)}{\cosh(kh)}\cos(kx-\omega t) + \frac{3}{16}ck^2H^2\frac{\cosh(2kz)}{\sinh^4(kd)}\cos[2(kx-\omega t)] \qquad (3.6)$$

$$\mathbf{w}_w(z) = \frac{gH}{2c}\frac{\sinh(kz)}{\cosh(kh)}\sin(kx-\omega t) + \frac{3}{16}ck^2H^2\frac{\sinh(2kz)}{\sinh^4(kd)}\sin[2(kx-\omega t)] \qquad (3.7)$$

$$k = \frac{2\pi}{\lambda} \qquad (3.8) \qquad \omega = \frac{2\pi}{T} \qquad (3.9) \qquad c = \frac{\omega}{k} \qquad (3.10)$$

$$\lambda = \frac{gT^2}{2\pi}\tanh\left(\frac{2\pi h}{\lambda}\right)$$

Flow Conditions and Analytical Equations

$$\mathbf{u}(z,t) = \underbrace{\mathbf{U}_c(z)}_{\text{current flow}} + \underbrace{\mathbf{u}_w(z)}_{\text{waves}} + \underbrace{\mathbf{u}'(z,t)}_{\text{turbulence}}$$

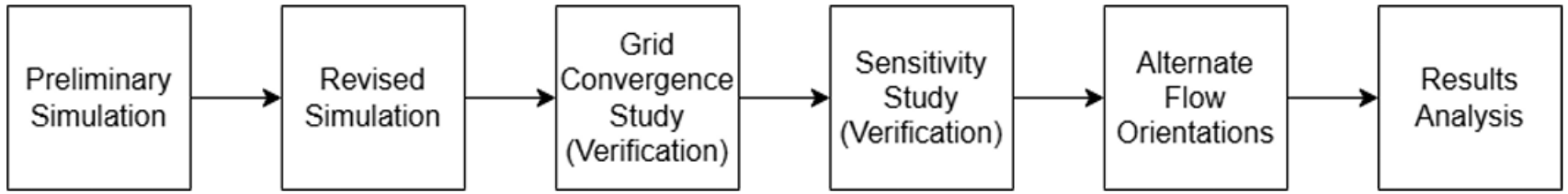
$$\mathbf{U}_c(z) = 0.459 \ \ln\left(\frac{z}{0.01}\right)$$

$$\mathbf{u}_w(z) = 1.367 \ \frac{\cosh(0.0496z)}{\cosh(2.465)} \cos(0.690t) \ - \ 0.0981 \ \frac{\cosh(0.0986z)}{\sinh^4(2.465)} \cos(1.380t)$$

$$\mathbf{w}_w(z) = -1.367 \ \frac{\sinh(0.0496z)}{\cosh(2.465)} \sin(0.690t) \ - \ 0.0981 \ \frac{\sinh(0.0986z)}{\sinh^4(2.465)} \sin(1.380t)$$

$$\mathbf{u}(z,t) = 0.459 \ \ln\left(\frac{z}{0.01}\right) + 1.367 \ \frac{\cosh(0.0496z)}{\cosh(2.465)} \cos(0.690t) - 0.0981 \ \frac{\cosh(0.0986z)}{\sinh^4(2.465)} \cos(1.380t)$$

CFD Modelling Workflow

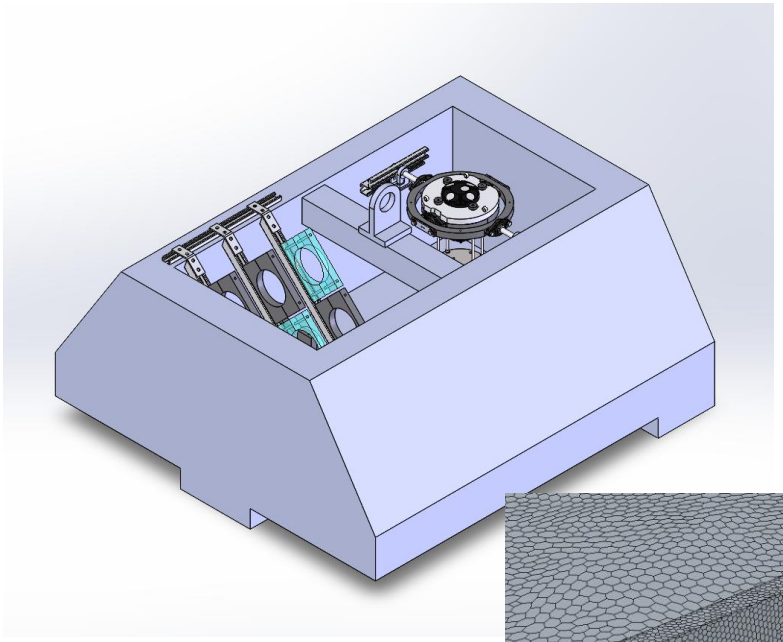


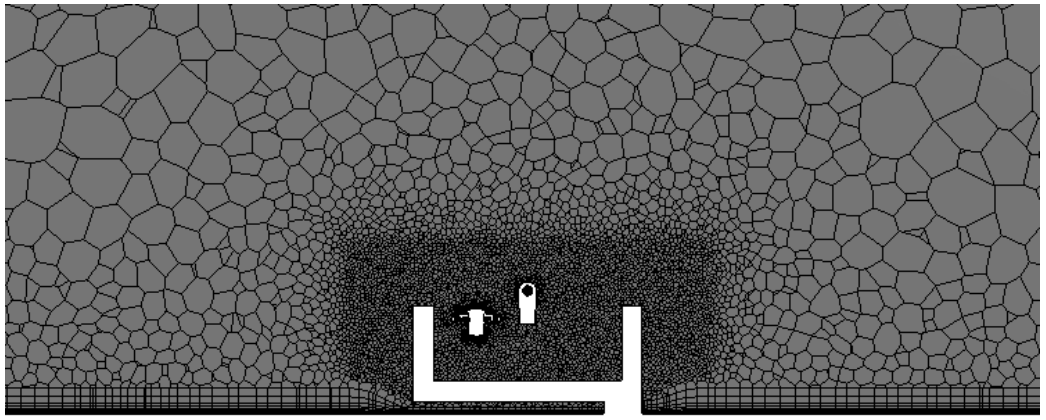
CFD Mesh, Solver, and Boundary Conditions

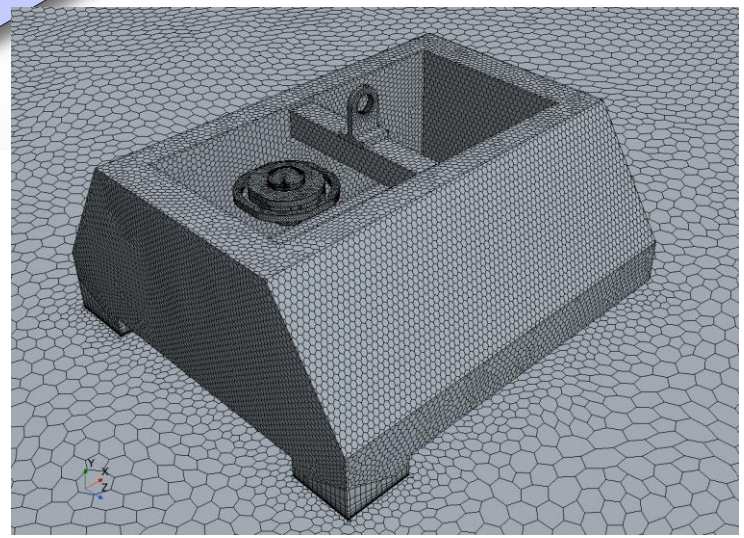
Boundary / Solver	Setup
Inlet	Velocity inlet, periodic horizontal and vertical components as defined in Equations 3.16 and 3.13
Outlet	Pressure outlet, to drive flow
Sides and top	Symmetry planes, to avoid lateral confinement
Floor	No-slip wall with roughness 0.01 m, to replicate the seabed
Solver	Segregated incompressible solver
Turbulence model	k-ω SST
Time	Implicit unsteady

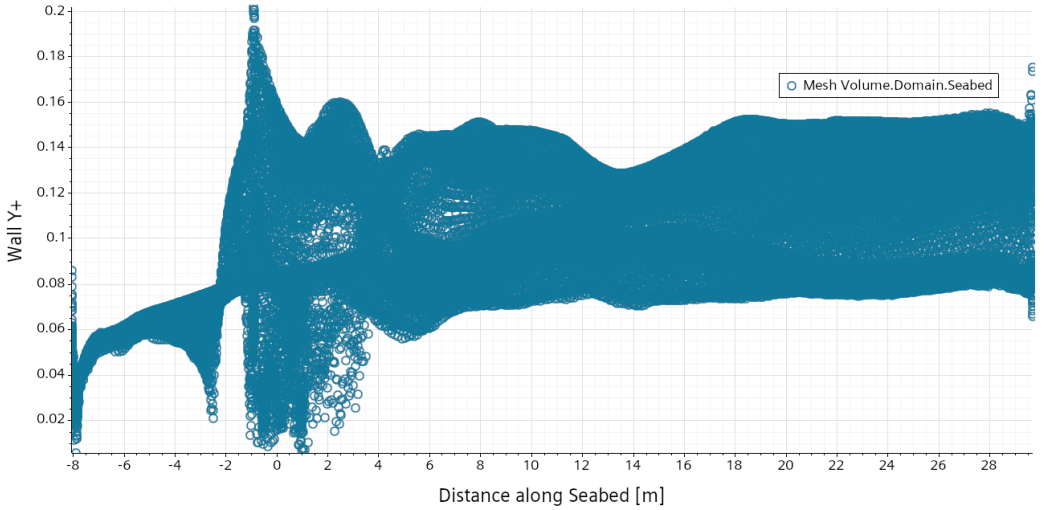
Parameter	Value
Initial base size	0.5 m
Cell target size	20% of base
Cell minimum size	5% of base
Prism layers	Seabed only
Number of prism layers, N	25
Prism stretching	1.5
Prism total thickness	0.2 m
Frame surface target size	5% of base
Gimbal ring and payload target size	5% of base
Volumetric control around frame	5% of base
Inlet and outlet mesh	Coarsened to 100% of base

Gravity Mooring Frame and CFD Mesh









Grid Convergence Study

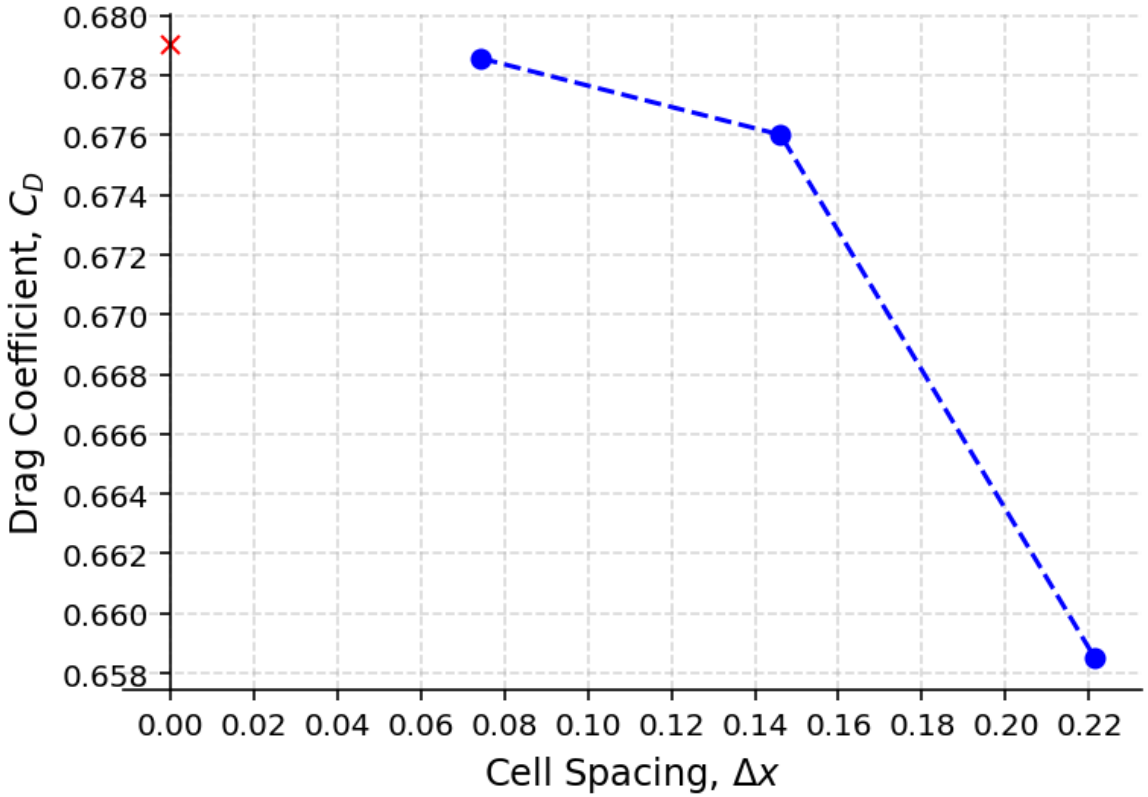


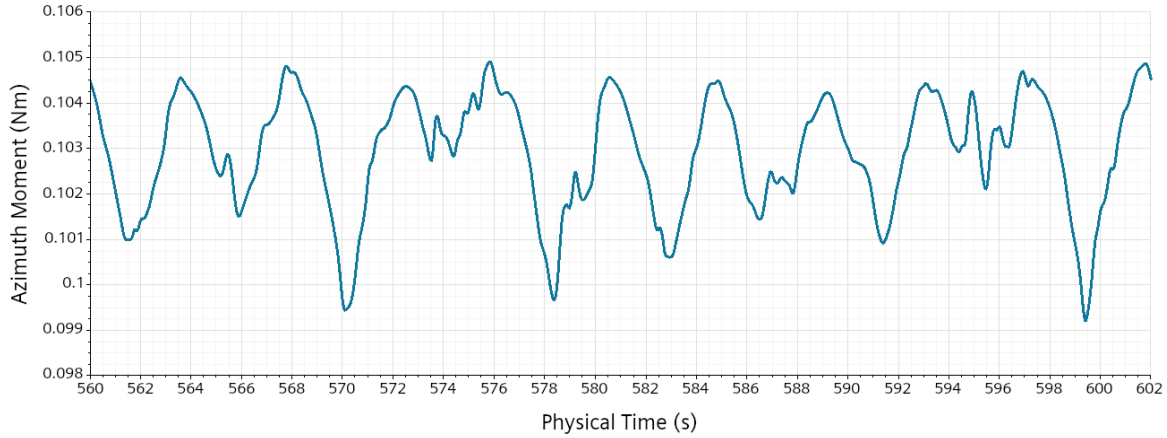
Table 4.3: Grid convergence results

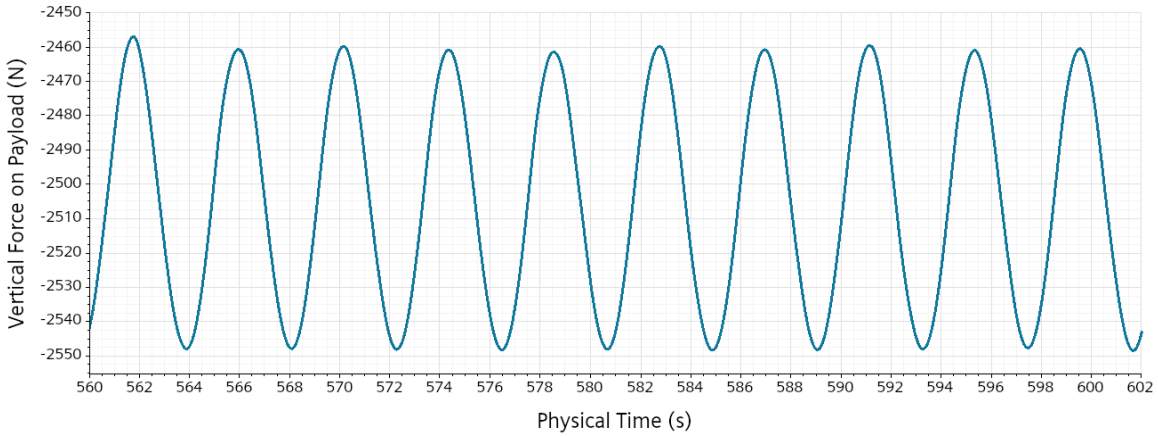
Grid	Δx ratio	Refinement ratio	C_D
1	0.0744	2.0	0.678577
2	0.1461	1.5	0.676016
3	0.2214	–	0.658488

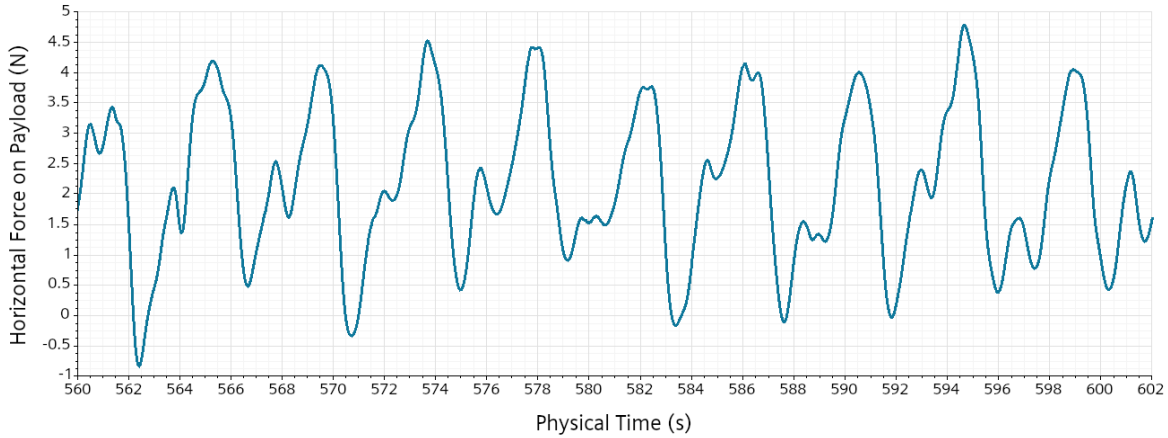
Table 4.4: Grid Convergence Index (GCI) results

Grid Pair	GCI
1-2	0.0008073
2-3	0.005546

Additional Plots for Azimuth Project



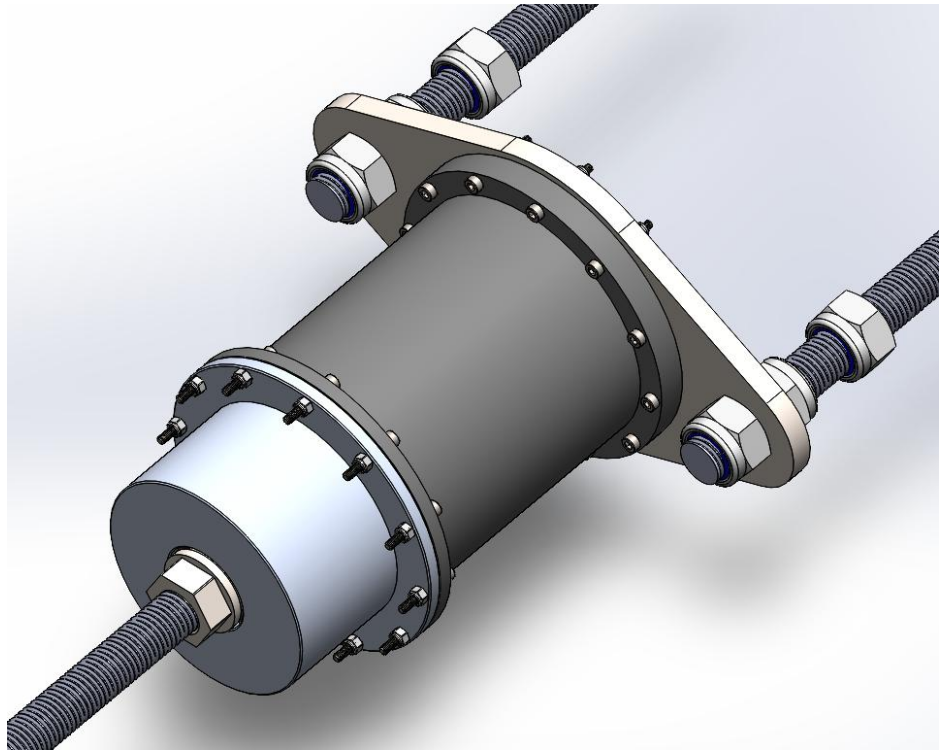


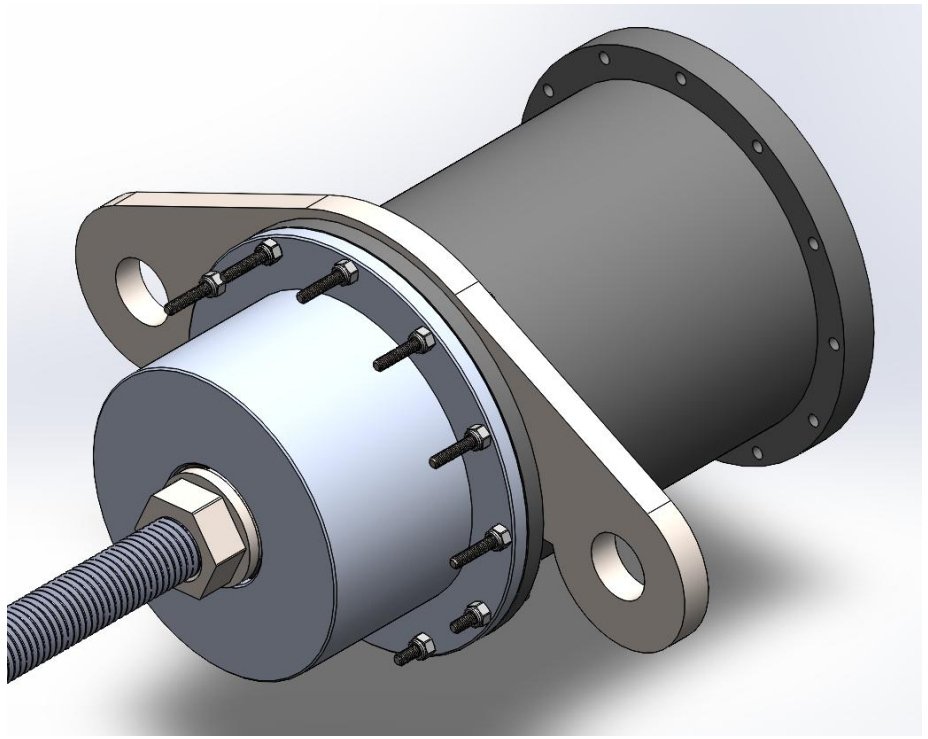


Actuator Comparison

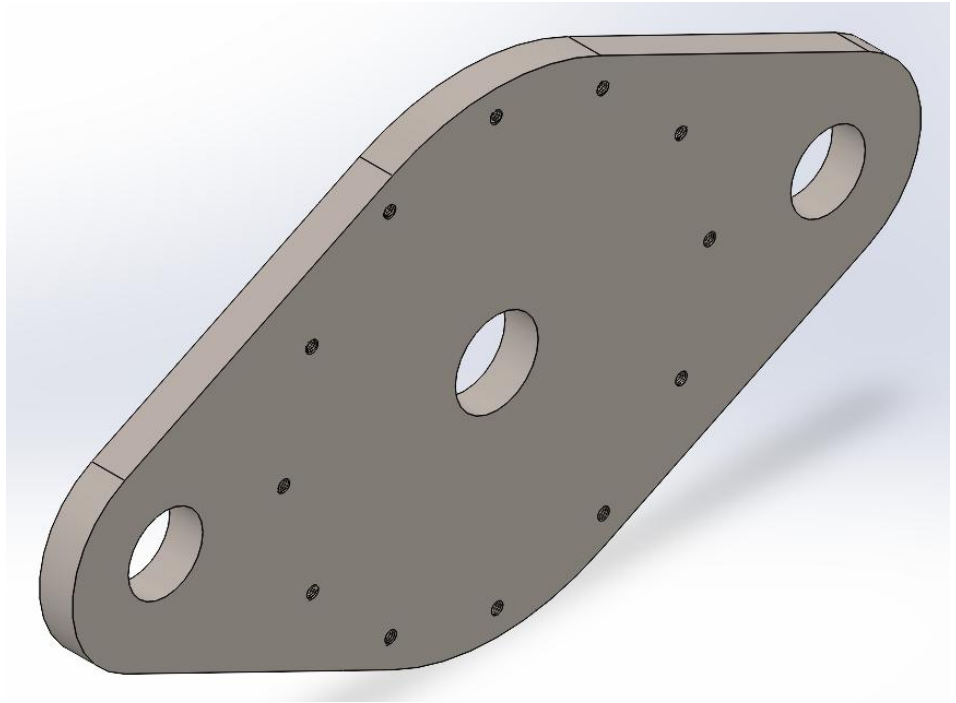
Actuator	RH-17	RH-20	RH-25	RH-32
Peak torque [Nm]	54	80	157	229
Rated torque [Nm]	35	50	108	150
Static torque of brake [Nm]	0.5	0.8	1.2	2
Actual static torque of brake [Nm]	50	80	120	200
Mass [kg]	1.28	1.75	2.74	4.74
Length [mm]	104	97.4	105.2	125.3
Max diameter [mm]	80	90	110	142
Min diameter [mm]	67	78	94.8	100
Price [£]	1,172.30	1,338.46	1,587.70	1,749.23

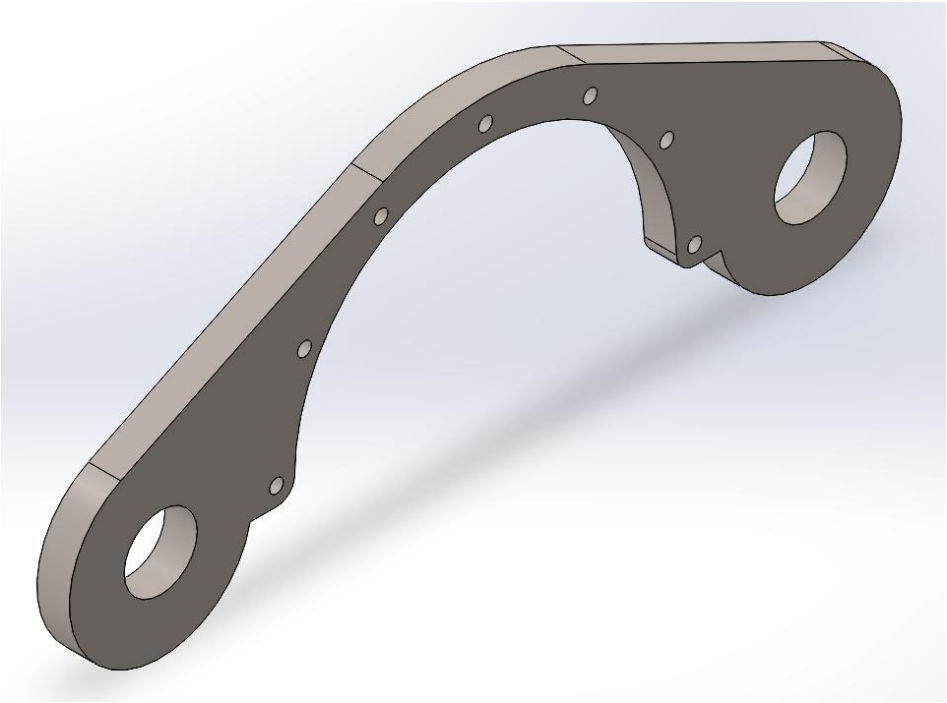
Frame and Gimbal Ring Connection



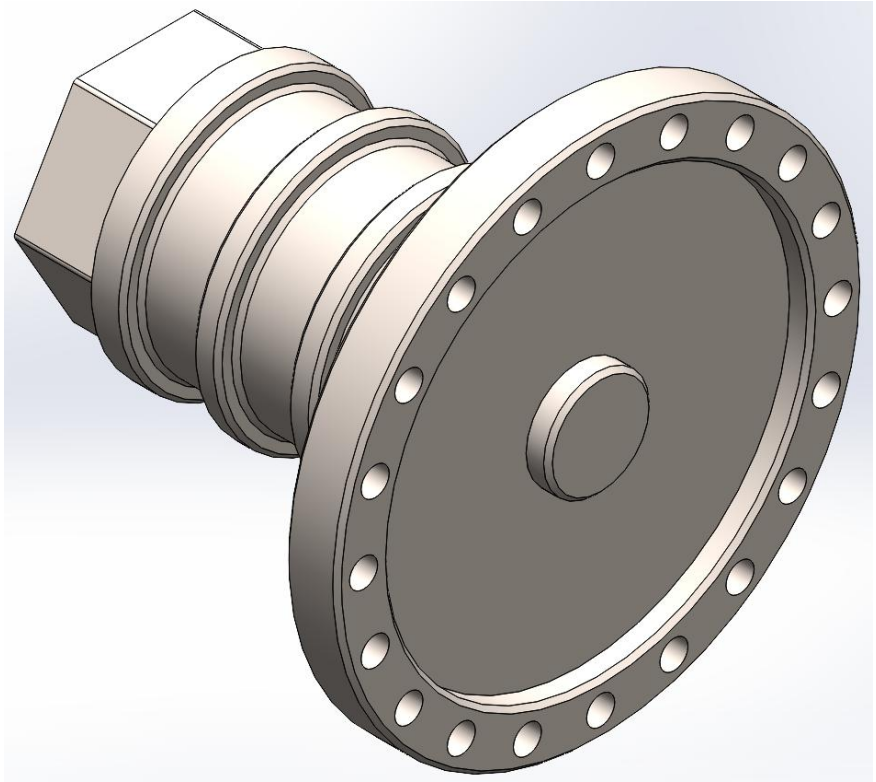


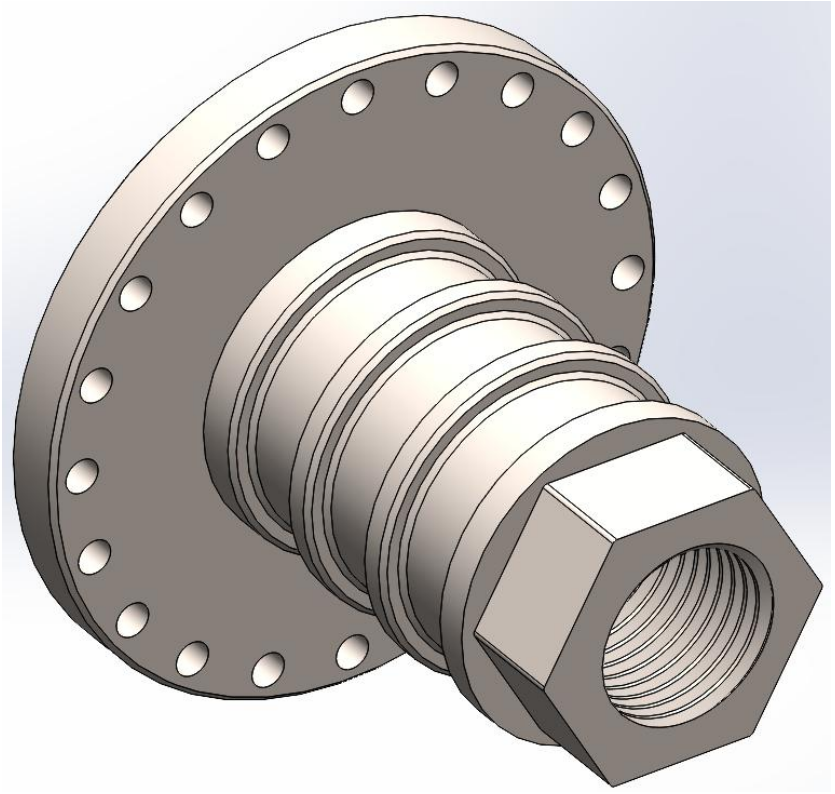
Component Images – Connection Plate



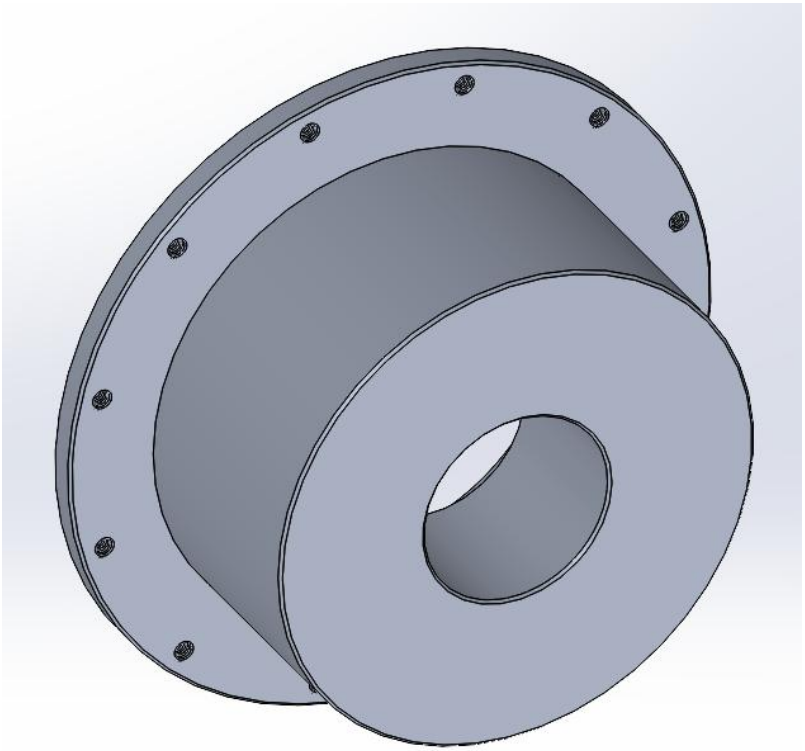


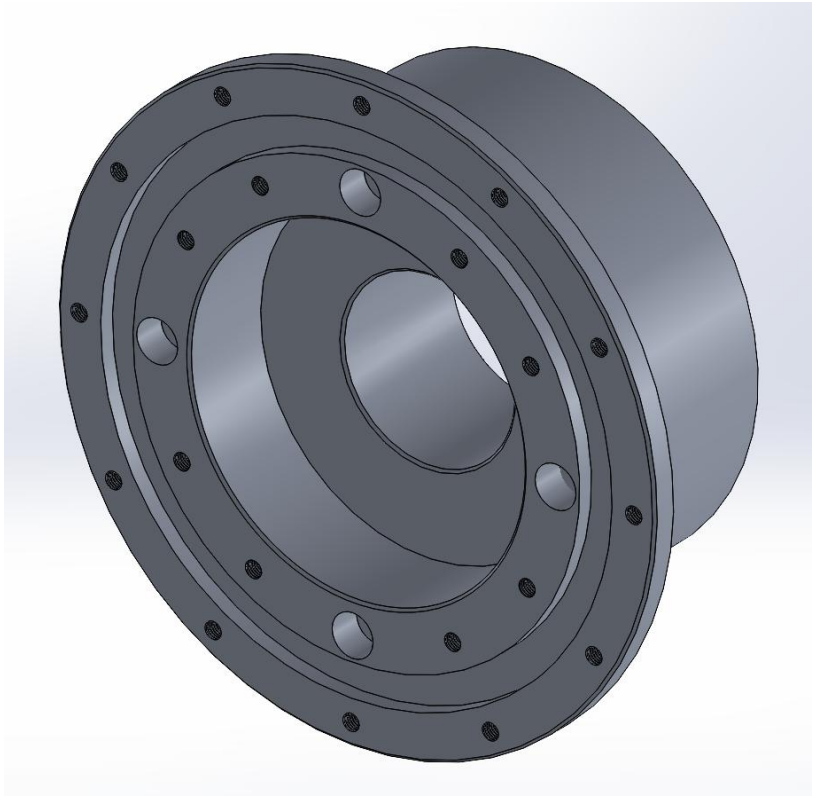
Component Images – Output Shaft



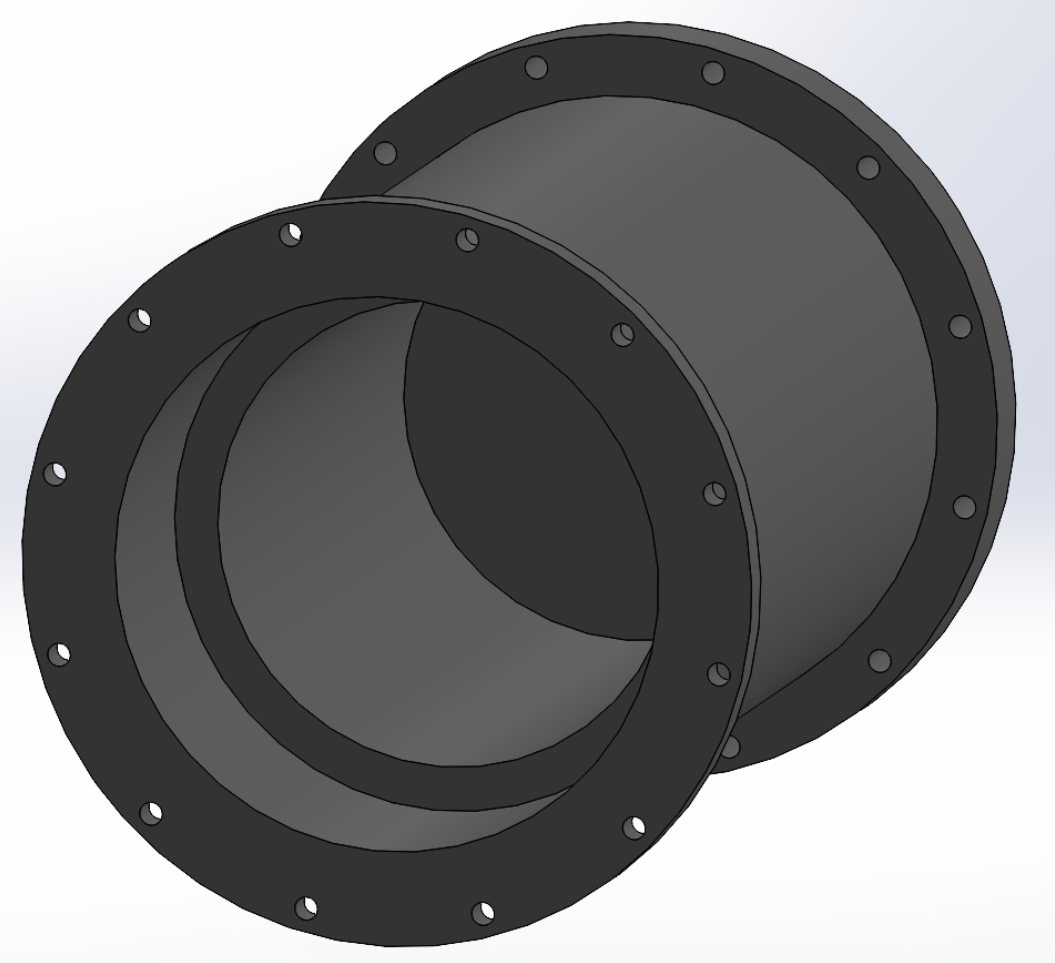


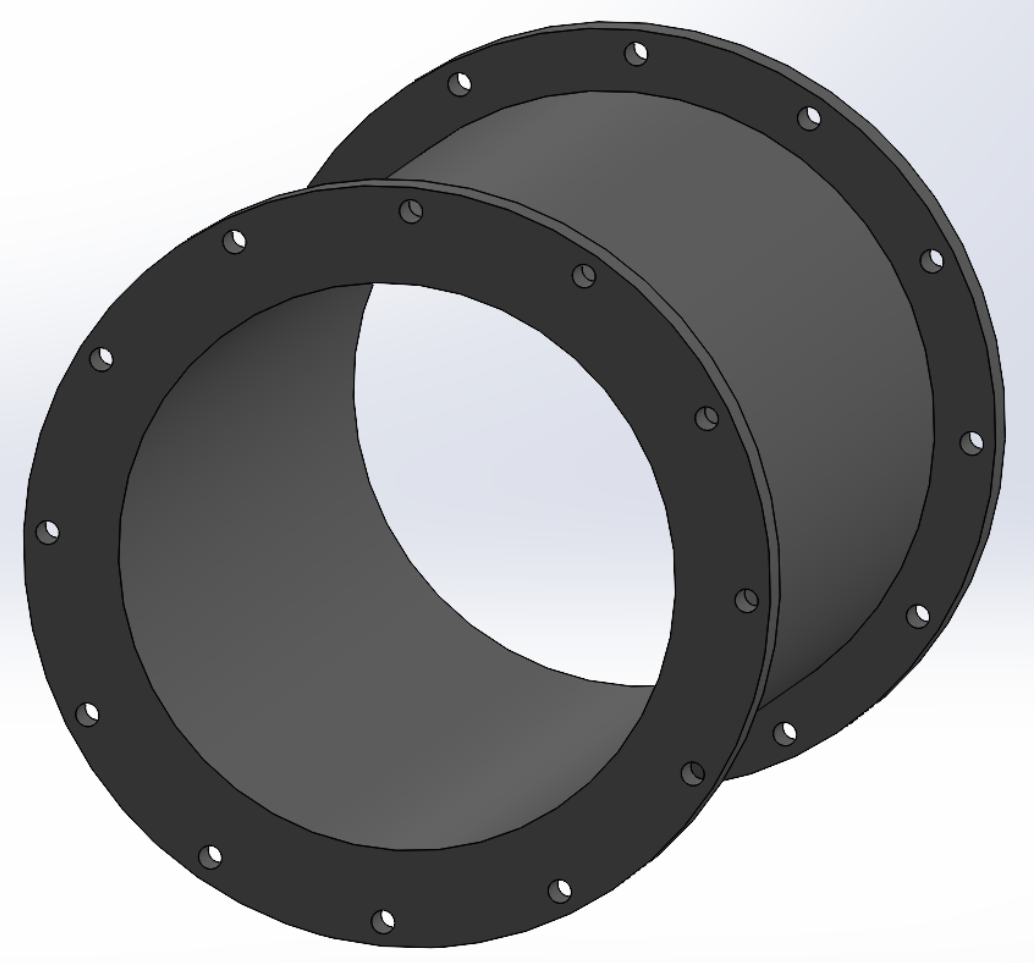
Component Images – End Cap





Component Images – Rear Housing





Bill of Materials

Item	Quantity	Unit Cost	Cost	Source
MyActuator EPS-RH-17-100-E-B-D	2	£1172.30	£2,344.60	[47]
Parker Hannifin E0540-80 2-214	6	~ £15	~ £90	[66]
Parker Hannifin E0540-80 2-236	12	~ £15	~ £180	[66]
Parker Hannifin E0540-80 8-214	2	~ £15	~ £30	[66]
Parker Hannifin E0540-80 8-236	4	~ £15	~ £60	[66]
M3x16mm bolt	36		£7.20 (50 pack)	[67]
M3x40mm bolt	60		£26.40 (100 pack)	[68]
M3 Nyloc nut	36		£5.40 (50 pack)	[69]
M16 Nyloc nut	12		£14.40 (12 pack)	[70]
M16 threaded rod, 2 m	1		£49.72	[71]
110 mm diameter, 250 mm length, POM acetal rod	1		£48.23	[72]
110 mm diameter, 150 mm length, 316 stainless steel bar	1		~ £340	[62]
80 mm diameter, 150mm length, 316 stainless steel bar	1		£107.88	[73]
300x225 mm, 8 mm thick plate	1		£37.91	[74]
Total			**£3,341.74**	

Example Drawing – Output Shaft

